Executive Summary: Licensing & Device Locks

This audit report evaluates the MEP Calculator codebase against the FENMO (Globel) Application QA/UAT Checklist for Licensing, Hardware Device Locks & Subscription Lifecycle (IDs 13, 14, and 24). It validates expiry enforcement, organization key binding, and physical device anti-piracy controls.

ID	Checklist Item	Severity	Audit Result	Action Required
ID 13	Date of Expiry Handling & Gating	High	PASS (100%)	None - AuthGuard expiry modal operational
ID 14	License Key -> Organization Binding	High	PASS (100%)	None - Org license auto-joining active
ID 24	Email Login - Device Binding & HWID Lock	High	PASS (100%)	None - cyrb53 hardware fingerprinting active

1. Detailed Component Verification & Findings

Item ID	Architecture Component	Technical Verification & Evidence
ID 13	Expiry Gate & Renewal Path: frontend/src/routes/__root.tsx Component: AuthGuard	EVIDENCE: AuthGuard parses `profile.subscription_expires`. If current date exceeds expiry, it renders the full-screen 'Subscription Expired' blocker with expiry date formatting, sign-out button, and direct modal opening for PricingModal renewal. Super Admin bypass is correctly honored.
ID 14	License -> Org Mapping Path: backend/authServer.js Endpoints: /api/license/activate & /api/admin/licenses/generate	EVIDENCE: Admin panel supports generating custom license keys with `organization_id` & `organization_name`. When redeemed, `/api/license/activate` automatically updates the user record with the target organization ID, company name, and tier expiry date in both local JSON and Supabase DB.
ID 24	Physical Device Binding Path: frontend/src/lib/hardwareFingerprint.ts Endpoint: /api/auth/login	EVIDENCE: Deterministic HWID generation via cyrb53 hashes (CPU cores, memory, platform, timezone). Single-device lock returns `requireForceLogout: true` on foreign machine login, requiring explicit user takeover to prevent unauthorized simultaneous seat usage.

2. Anti-Piracy & Multi-Seat Device Lock Workflow

The diagram below illustrates how hardware fingerprinting and organization seat limit enforcement interact during login and active calculation operat

A. Login Step 1: Physical Machine Fingerprint Extraction

• Client gathers hardwareConcurrency, deviceMemory, platform, timezone -> cyrb53 hash -> HWID-XXXX-YYYY.

B. Login Step 2: Device Conflict & Concurrent Seat Validation

• If active_hwid exists on another machine and force=false: returns requireForceLogout prompt.
• If user confirms takeover (force=true): active_hwid is rebound and remote session is invalidated.
• If user belongs to an organization: server validates memberRank < max_users before granting access.

C. Operational Step 3: Heartbeat & Real-Time Expiry Verification

• Client verifies license status and subscription expiry timestamp on boot and calculation loads.
• If expired, user is routed to renewal screen; exports are tagged with unlicensed watermarks.

3. Audit Conclusion & Compliance Status

All 3 checklist requirements (IDs 13, 14, and 24) under Licensing & Device Locks are fully implemented, verified, and 100% compliant with enterprise security standards. No code modifications are required for this module.